Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 1 / 方法 1 [CO02-M01]

Subsets are unordered choices. Translate each condition into forced membership, a size split, or a complement before expanding formulas. Binomial identities become reliable when both sides count the same objects, while modular coefficient arguments require a clearly specified polynomial ring.

子集是无序选取。展开公式前，先把条件转成强制归属、大小分类或补集。二项式恒等式应由两边计算同一对象来保证，模系数论证则须明确使用的多项式环。

Condition a selection on its type counts

按类型数量分类选取

Procedure

1. Choose a size or type-count vector whose cases are disjoint and exhaustive.

2. Multiply the binomial choices from disjoint groups, including any forced or excluded elements.

3. Sum the valid vectors; when solving for an unknown size, cancel factorial ratios before testing its integer domain.

通用步骤

1. 选择互斥且完备的大小或类型数量向量。

2. 相乘各不交组的组合选法，并处理必选或禁选元素。

3. 对可行向量求和；未知总大小时先约去阶乘比例，再在整数定义域内求解。

> Example. Choose four books from six novels and four poetry books. Both genres must appear, and the number of novels must be even. Count the choices.

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 1 / 方法 1 [CO02-M01] (continued / 续)

> 例题：从六本小说、四本诗集中选四本，要求两类均出现且小说数量为偶数，求选法数。

Worked solution

1. The novel count can only be two: zero or four would omit a genre.

2. Choose two novels and two poetry books independently.

3. The count is $\binom{6}{2}\binom{4}{2} = 15 \cdot 6 = 90$. No ordering factor is needed.

例题解析

1. 小说数只能为二，零或四都使一类缺席。

2. 独立选择两本小说与两本诗集。

3. 共有 $\binom{6}{2}\binom{4}{2} = 15 \cdot 6 = 90$ 种，不需要排列因子。

Choose the sample space once: subsets, ordered draws, and assignments to labeled people have different denominators.

先固定样本空间：子集、有序抽取与向有标识人分配具有不同分母。

Canceling $\binom{n}{r}/\binom{n}{s}$ often leaves a short product of consecutive factors; prove its monotonicity on the legal range before declaring an integer solution unique. For boundary-point triangles, subtract triples lying on a common side from all triples, and justify that no other collinear triple exists.

约去 $\binom{n}{r}/\binom{n}{s}$ 的阶乘常剩连续因子的短乘积；须在合法范围证明单调后才可断言整数解唯一。对边界点三角形，从全部三元组扣除同边共线三元组，并证明没有其他共线三点。

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 1 / 方法 1 [CO02-M01] (continued / 续)

Practice / 配套练习: AIME-2000-II-Q03 (WB p. 720), AIME-2004-II-Q02 (WB p. 749), AIME-2005-I-Q03 (WB p. 601), AIME-2005-II-Q01 (WB p. 629), AIME-2006-II-Q04 (WB p. 616), AIME-2007-I-Q10 (WB p. 634), AIME-2009-I-Q11 (WB p. 254), AIME-2010-II-Q13 (WB p. 764), AIME-2011-I-Q10 (WB p. 739), AIME-2012-II-Q03 (WB p. 630), AIME-2013-I-Q06 (WB p. 740), AIME-2015-I-Q02 (WB p. 726), AIME-2017-I-Q01 (WB p. 631), AIME-2018-I-Q09 (WB p. 664), AIME-2018-I-Q10 (WB p. 715), AIME-2022-II-Q09 (WB p. 700), AIME-2023-I-Q01 (WB p. 731), AIME-2024-I-Q04 (WB p. 760), AIME-2025-I-Q05 (WB p. 495), AIME-2025-I-Q10 (WB p. 627)

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 2 / 方法 2 [CO02-M02]

Encode subsets by membership states and intrinsic levels

用归属状态与内在层次编码子集

Procedure

1. Describe each ground-set element by which sets contain it; translate global conditions into allowed states or fixed atoms.

2. Choose elements for any distinguished atoms and freely assign all remaining allowed states.

3. If the object is a function, choose intrinsic image or preimage levels first, then count maps between those levels.

4. Check that the encoding can be recovered from the object; otherwise different state descriptions may be duplicates.

通用步骤

1. 按每个基础元素属于哪些集合来编码，将整体条件转为允许状态或固定区域。

2. 为特殊区域选择元素，再自由分配其余允许状态。

3. 若对象为函数，先选其内在像集或原像层，再计层间映射。

4. 检验对象能否恢复编码，否则不同状态描述可能重复。

> Example. For three subsets of a five-element set, exactly two elements belong to all three, and each other element belongs to at most one. Count the ordered triples of subsets.

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 2 / 方法 2 [CO02-M02] (continued / 续)

例题：五元集合的三个子集中，恰有两个元素同时属于三者，其余每个元素至多属于一个子集，求有序子集三元组数。

Worked solution

1. Choose the common two-element atom in $\binom{5}{2} = 10$ ways.

2. Each of the other three elements has four states: only in the first, only in the second, only in the third, or in none.

3. Every triple uniquely recovers its common atom and states, so the count is $10 \cdot 4^3 = 640$.

例题解析

1. 选择三者公共的二元区域有 $\binom{5}{2} = 10$ 种。

2. 余下三个元素各有仅属第一、仅属第二、仅属第三、均不属四种状态。

3. 每个集合三元组唯一恢复公共区域与各状态，故为 $10 \cdot 4^3 = 640$。

The empty subset is one legal state unless it is explicitly excluded.

除非明确排除，空子集本身也是一个合法状态。

Exactly 2^{a-1} nonempty finite subsets of positive integers have maximum a: include a and choose freely below it. Therefore a sum of such counts can be decoded uniquely by binary expansion.

For $f \circ f$ constant with value a, the intrinsic level $S = f^{-1}(a)$ contains the fixed point a; points outside S must map into $S \setminus \{a\}$. For $f^2 = f^3$, use fixed points, their nonfixed direct preimages, and the remaining level. The levels must be intrinsic to avoid counting the same function more than once.

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 2 / 方法 2 [CO02-M02] (continued / 续)

最大元恰为 a 的有限非空正整数子集有 2^{a-1} 个：必选 a，其下任意选。因此此类数量之和可以按二进制展开唯一反解。

若 $f \circ f$ 恒为 a，内在层 $S = f^{-1}(a)$ 含不动点 a，S 外元素必须映入 $S \setminus \{a\}$。若 $f^2 = f^3$，则分为不动点、直接映入不动点的非不动元素以及剩余层。各层必须由函数本身唯一确定，以免重复计数。

Practice / 配套练习: AIME-2002-II-Q09 (WB p. 633), AIME-2010-I-Q07 (WB p. 636), AIME-2010-II-Q08 (WB p. 637), AIME-2013-II-Q11 (WB p. 215), AIME-2017-II-Q01 (WB p. 660), AIME-2018-II-Q10 (WB p. 207), AIME-2021-II-Q06 (WB p. 640), AIME-2024-II-Q06 (WB p. 642)

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 3 / 方法 3 [CO02-M03]

Collapse binomial sums by symmetry and convolution

用对称与卷积合并二项式和

Procedure

1. Rewrite every factor so the lower indices total a fixed selection size.

2. Interpret a convolution by choosing from disjoint groups: $\sum_k \binom{a}{k}\binom{b}{r-k} = \binom{a+b}{r}$.

3. Use complementary indices to obtain half-row or squared-row sums; retain any middle coefficient when the row length is even.

4. For signed or residue weights, use generating functions or an explicit parity filter rather than discarding signs.

通用步骤

1. 改写各因子，使下标之和成为固定选取数。

2. 把卷积解释为从不交组中选取：$\sum_k \binom{a}{k}\binom{b}{r-k} = \binom{a+b}{r}$。

3. 用互补下标处理半行和或平方和；行长为偶数时保留中间系数修正。

4. 有符号或剩余类权重时，使用生成函数或明确奇偶筛选，不能忽略符号。

> Example. From six red and six blue objects, choose six. How many choices contain an even number of red objects?

> 例题：从六红六蓝物体中选六个，含偶数个红物体的选法有多少？

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 3 / 方法 3 [CO02-M03] (continued / 续)

Worked solution

1. Let E, O count choices with even and odd red counts. Vandermonde gives $E + O = \binom{12}{6} = 924$.

2. The signed count $E - O$ is the coefficient of x^6 in $(1-x)^6(1+x)^6 = (1-x^2)^6$, namely $-\binom{6}{3} = -20$.

3. Adding the equations gives $E = (924 - 20)/2 = 452$.

例题解析

1. 令 E, O 分别计偶数与奇数个红物体，范德蒙德给出 $E + O = \binom{12}{6} = 924$。

2. 带符号差 $E - O$ 是 $(1-x)^6(1+x)^6 = (1-x^2)^6$ 的 x^6 系数，即 $-\binom{6}{3} = -20$。

3. 两式相加得 $E = (924 - 20)/2 = 452$。

A complement changes an index but does not change a binomial coefficient: $\binom{n}{k} = \binom{n}{n-k}$.

取补改变下标但不改变组合数：$\binom{n}{k} = \binom{n}{n-k}$。

Practice / 配套练习: AIME-2000-II-Q07 (WB p. 632), AIME-2003-I-Q13 (WB p. 591), AIME-2006-II-Q10 (WB p. 785), AIME-2007-I-Q10 (WB p. 634), AIME-2007-II-Q10 (WB p. 738), AIME-2008-II-Q12 (WB p. 654), AIME-2017-I-Q07 (WB p. 638), AIME-2018-I-Q12 (WB p. 748), AIME-2020-I-Q07 (WB p. 639), AIME-2022-I-Q12 (WB p. 668), AIME-2025-I-Q10 (WB p. 627)

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 4 / 方法 4 [CO02-M04]

Count weakly ordered indices and telescope binomial rows

计算弱次序下标并望远镜合并二项式行

Procedure

1. Translate inverse-branch choices or compressed positions into a weakly increasing index list.

2. Add successive offsets to turn weak inequalities into a strict subset selection.

3. When summing over the last index, use Pascal differences or the hockey-stick identity, with the starting index checked.

通用步骤

1. 把反函数分支或压缩位置转为弱递增下标串。

2. 依次加偏移量，将弱不等式转为严格子集选取。

3. 按最后下标求和时，用帕斯卡差分或曲棍球杆恒等式，并核对起始下标。

> Example. Count triples of days $2 \le i \le j \le k \le 7$, then count those whose last day is exactly seven.

> 例题：求 $2 \le i \le j \le k \le 7$ 的日期三元组数，并求末日恰为七的三元组数。

Worked solution

1. The map $(i, j, k) \mapsto (i, j+1, k+2)$ gives three distinct choices from $2, \ldots, 9$, so the total is $\binom{8}{3} = 56$.

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 4 / 方法 4 [CO02-M04] (continued / 续)

2. If $k = 7$, choose a weak pair $2 \le i \le j \le 7$, equivalent to two distinct elements from $2, \ldots, 8$.

3. This gives $\binom{7}{2} = 21$, also equal to $\binom{8}{3} - \binom{7}{3}$.

例题解析

1. 映射 $(i, j, k) \mapsto (i, j+1, k+2)$ 给出从二至九选三个不同数，总数为 $\binom{8}{3} = 56$。

2. 若 $k = 7$，只需选弱递增二元组 $2 \le i \le j \le 7$，等价于从二至八选两个不同数。

3. 故为 $\binom{7}{2} = 21$，也等于 $\binom{8}{3} - \binom{7}{3}$。

A half-open analytic branch can impose a strict upper endpoint even when the index inequality is weak.

解析分支若为半开区间，即使下标弱递增，也须保留严格上端点。

Practice / 配套练习: AIME-2009-II-Q06 (WB p. 635), AIME-2020-II-Q14 (WB p. 178), AIME-2022-II-Q10 (WB p. 641), AIME-2023-I-Q11 (WB p. 687)

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 5 / 方法 5 [CO02-M05]

Propagate binomial weights modulo a prime

在素数模下传递二项式权重

Procedure

1. Repeated adjacent addition produces Pascal weights; write the coefficient of each initial entry.

2. Use $(1+x)^p \equiv 1 + x^p \pmod{p}$ to reduce a large exponent.

3. For sums over exponents modulo m, work also modulo $x^m - 1$, where those coefficients combine.

4. Check which remaining factorial coefficients are nonzero modulo the prime before counting zero classes.

通用步骤

1. 反复相邻相加产生帕斯卡权重，写出每个初始数的系数。

2. 用 $(1+x)^p \equiv 1 + x^p \pmod{p}$ 降低大指数。

3. 若按指数模 m 求和，还须模去 $x^m - 1$，相应系数才会合并。

4. 计算哪些剩余阶乘系数在素数模下非零，再数零剩余类。

> Example. Find the coefficient of x^6 in $(1+x)^{13}$ modulo five.

> 例题：求 $(1+x)^{13}$ 的 x^6 系数模五的值。

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 5 / 方法 5 [CO02-M05] (continued / 续)

Worked solution

1. In characteristic five, $(1+x)^5 \equiv 1 + x^5$, so $(1+x)^{13} \equiv (1+x)^3(1+x^5)^2$.

2. To obtain degree six, take x from the first factor and x^5 from the second; no other exponents can combine to six.

3. The coefficient is $\binom{3}{1}\binom{2}{1} = 6 \equiv 1 \pmod 5$.

例题解析

1. 模五时 $(1+x)^5 \equiv 1 + x^5$，所以 $(1+x)^{13} \equiv (1+x)^3(1+x^5)^2$。

2. 次数六只能由第一因子的一次项与第二因子的五次项相乘。

3. 系数为 $\binom{3}{1}\binom{2}{1} = 6 \equiv 1 \pmod 5$。

Reducing exponents modulo $p-1$ applies in the quotient by $x^{p-1}-1$, not as an identity of ordinary polynomials.

指数模 $p-1$ 约化发生在模 $x^{p-1}-1$ 的商环中，不是普通多项式恒等式。

Practice / 配套练习: AIME-2007-II-Q13 (WB p. 643), AIME-2026-I-Q13 (WB p. 646)

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 6 / 方法 6 [CO02-M06]

Rank fixed-weight binary strings by prefix blocks

按前缀区块确定定重二进制串排名

Procedure

1. Fix the length and required number of ones, including a leading-digit rule if present.

2. At a possible one, count the smaller zero-prefix block by a binomial coefficient.

3. For ranking, add all skipped block sizes; for unranking, subtract a block only when the target rank lies beyond it.

4. For consecutive equal-weight integers, analyze the terminal carry chain before counting the shared prefix.

通用步骤

1. 固定长度与一的总数，并写明是否允许前导零。

2. 遇到可选的一位时，用组合数计算该位为零的较小前缀块。

3. 求排名时相加所有跳过区块；反解排名时，仅当目标越过区块才减去其大小。

4. 对相邻且同重量整数，先分析末尾进位链，再计算公共前缀。

> Example. Among length-six binary strings with three ones, including initial zeros and ordered as integers, what is the rank of 101010?

> 例题：六位二进制串恰有三个一，允许前导零，按整数大小排列。101010 排第几?

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 6 / 方法 6 [CO02-M06] (continued / 续)

Worked solution

1. Before its initial one are strings beginning zero, giving $\binom{5}{3} = 10$.

2. At the third bit, the smaller prefix 100 leaves two ones in three places, giving $\binom{3}{2} = 3$. At the fifth bit, prefix 10100 leaves one one in one place, giving one more.

3. There are $10 + 3 + 1 = 14$ smaller strings, so the rank is 15.

例题解析

1. 首位一之前是以零开头的串，有 $\binom{5}{3} = 10$ 个。

2. 第三位的一之前，较小前缀 100 余三位放两个一，有 $\binom{3}{2} = 3$ 个；第五位之前，前缀 10100 余一位放一个一，再有一个。

3. 较小串共 $10 + 3 + 1 = 14$ 个，排名为十五。

Rank one is the first object; count smaller objects and then add one.

排名从一开始：先计更小对象，再加一。

Practice / 配套练习: AIME-2012-I-Q05 (WB p. 569), AIME-2012-II-Q07 (WB p. 587)

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 7 / 方法 7 [CO02-M07]

Classify a forbidden-subset family before selecting members

先分类禁用子集族结构再选成员

Procedure

1. Find a distinguished member or minimum element that forces exclusions.

2. Convert pairwise restrictions into an auxiliary graph when members are edges or small subsets.

3. Prove the structural list is complete, then choose freely only within the surviving independent components.

通用步骤

1. 找出能迫使排除条件的特殊成员或最小元素。

2. 若成员为边或小子集，将两两限制转成辅助图。

3. 证明结构分类完备，再仅在剩余独立部分自由选取。

> Example. Prove that four distinct edges of a simple graph which pairwise intersect must all share one vertex.

> 例题：证明简单图中四条不同的边若两两相交，则必共用一个顶点。

Worked solution

1. Take two edges ab, ac. If every remaining edge contains a, the conclusion holds.

2. Any edge avoiding a must intersect both, so it is bc.

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 7 / 方法 7 [CO02-M07] (continued / 续)

3. An edge meeting all of ab, ac, bc can only be one of these three unless it contains a common vertex of all three, which does not exist. Thus a fourth distinct edge rules out the triangle case, leaving a star.

例题解析

1. 取两边 ab, ac，若其余边均含 a，结论成立。

2. 若某边不含 a，为了与这两边相交，只能是 bc。

3. 能同时交 ab, ac, bc 的边只能是这三条，因为三者没有公共顶点。所以第四条不同边排除了三角形情况，只剩星形。

Pairwise intersecting does not mean globally intersecting: a triangle is the essential exceptional case.

两两相交不等于全部有公共交点，三角形是关键例外。

For a pairwise-intersecting family of exactly half the subsets of a finite set, complementary pairs allow at most one member, hence exactly one from every pair. A chosen singleton forces all members to contain that element. Without a singleton, analyze the selected two-subsets as an intersecting edge family: empty, star subfamily, or triangle.

For product-free subsets, condition on the least chosen integer. Its products create explicit forbidden pairs or triples among the remaining integers. Count independent components only after verifying all product relations have been covered, including a repeated factor such as a^2.

两两相交子集族若恰含全部子集的一半，每对互补集至多选一个，因此每对恰选一个。若选单点集，全部成员都必须含该元素；没有单点集时，将选中的二元集视为两两相交边族，只可能为空、某星形的子族或三角形。

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 7 / 方法 7 [CO02-M07] (continued / 续)

对无乘积关系的子集，按最小所选整数分类，它与其他数相乘会产生明确的禁用二元或三元组合。只有确认覆盖全部乘积关系，包括 a^2 这类重复因子后，才可按独立分量相乘。

Practice / 配套练习: AIME-2017-I-Q12 (WB p. 644), AIME-2023-II-Q11 (WB p. 645)